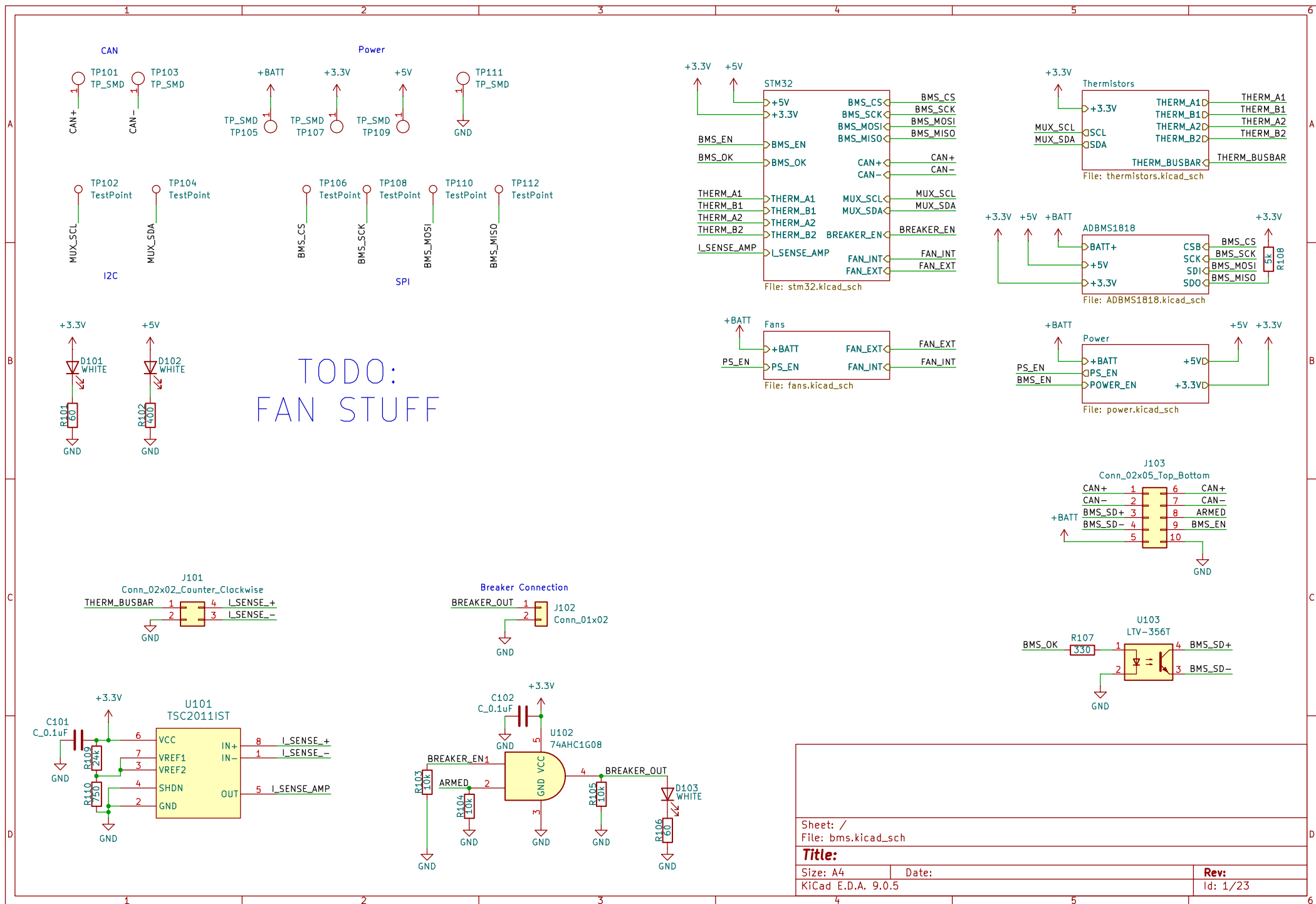




This is a Hierarchical Sheet.
That means that this is a "copy-paste"
block for each block shown.
(ie. As seen on "Voltage Filter")

"A cell pin filter of 100 Ω and 10 nF
is recommended for all applications"
ADBMS1818 datasheet pg. 71

However, since it limits the balancing current...

"Note that the antialiasing filter resistor is part of the discharge path
and must be removed or reduced. Use of an RC for added cell
voltage measurement filtering is permitted, but the filter resistor
must remain small, typically around 10 Ω to reduce the effect on the
balance current." pg. 72

Filtering RC circuit as depicted on
ADBMS1818 datasheet pg. 72

Balance Current

$V = 4.2 \text{ V}$

$R = R_{\text{DISCHARGE}} + R_{\text{FILTER}} + R_{\text{DS_ON}}$
 $= 26 \Omega + 10 \Omega + 4 \Omega$
 $= 40 \Omega$

$I = 4.2 \text{ V} / 40 \Omega$
 $= 105 \text{ mA}$

Power

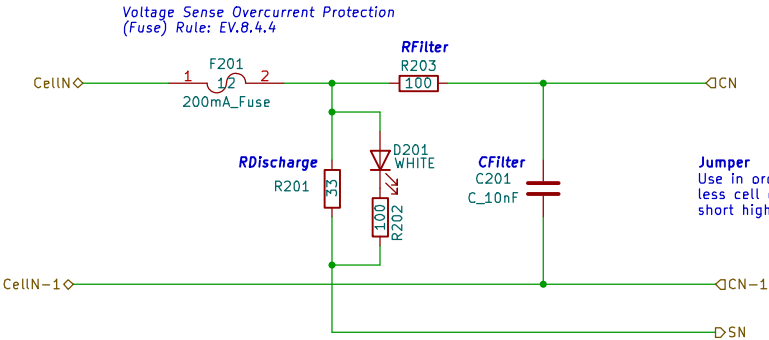
$P = I^2 R$
 $= (105 \text{ mA})^2 * 40 \Omega$
 $= 0.441 \text{ W}$

Balance current [A]

$= (\% \text{ SOC imbalance} * \text{battery capacity}) / (\text{num of hours to balance})$
 $= (0.05 * 64\text{Ah}) / (20)$
 $= 160 \text{ mA}$

Discharge Resistor [ohms]

$= \text{cell voltage} / \text{balancing current}$
 $= 4.2 \text{ V} / 160\text{mA}$
 $= 26.25 \text{ ohms}$



As long as balance current is below 200mA,
everything is okay to use the internal MOSFETs
of the ADBMS1818 chip for balancing.

This is the calculated balance current for the 10087191NSH5
cells, trying to keep it under 200mA. The example in the
ADBMS1818 datasheet uses 5% SOC imbalance being resolved
in 5 hours, but we can push to a longer balancing time if needed.

Sheet: /ADBMS1818/Cell1 Filtering and Balancing/
File: filter_balance.kicad_sch

Title:

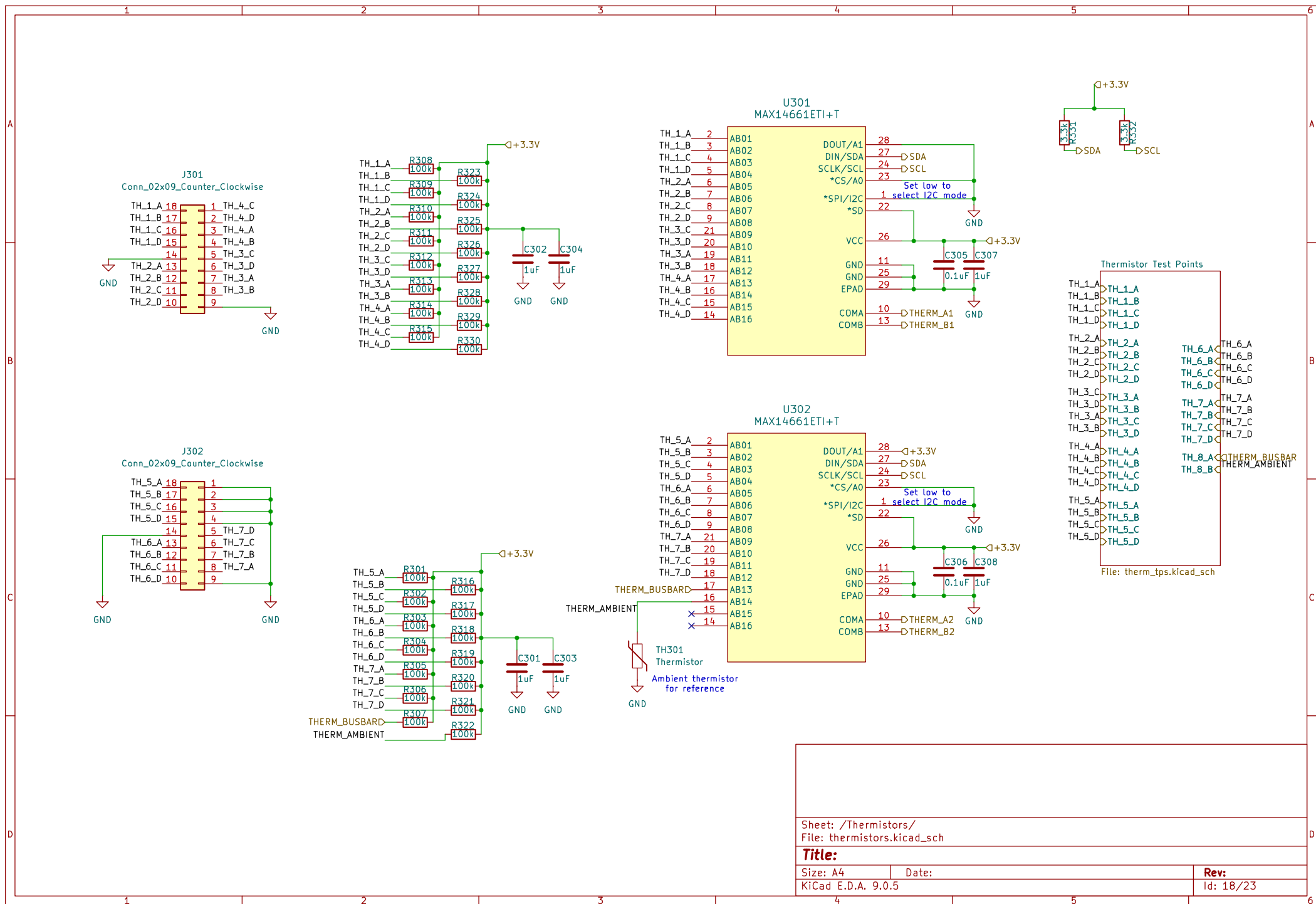
Size: A4

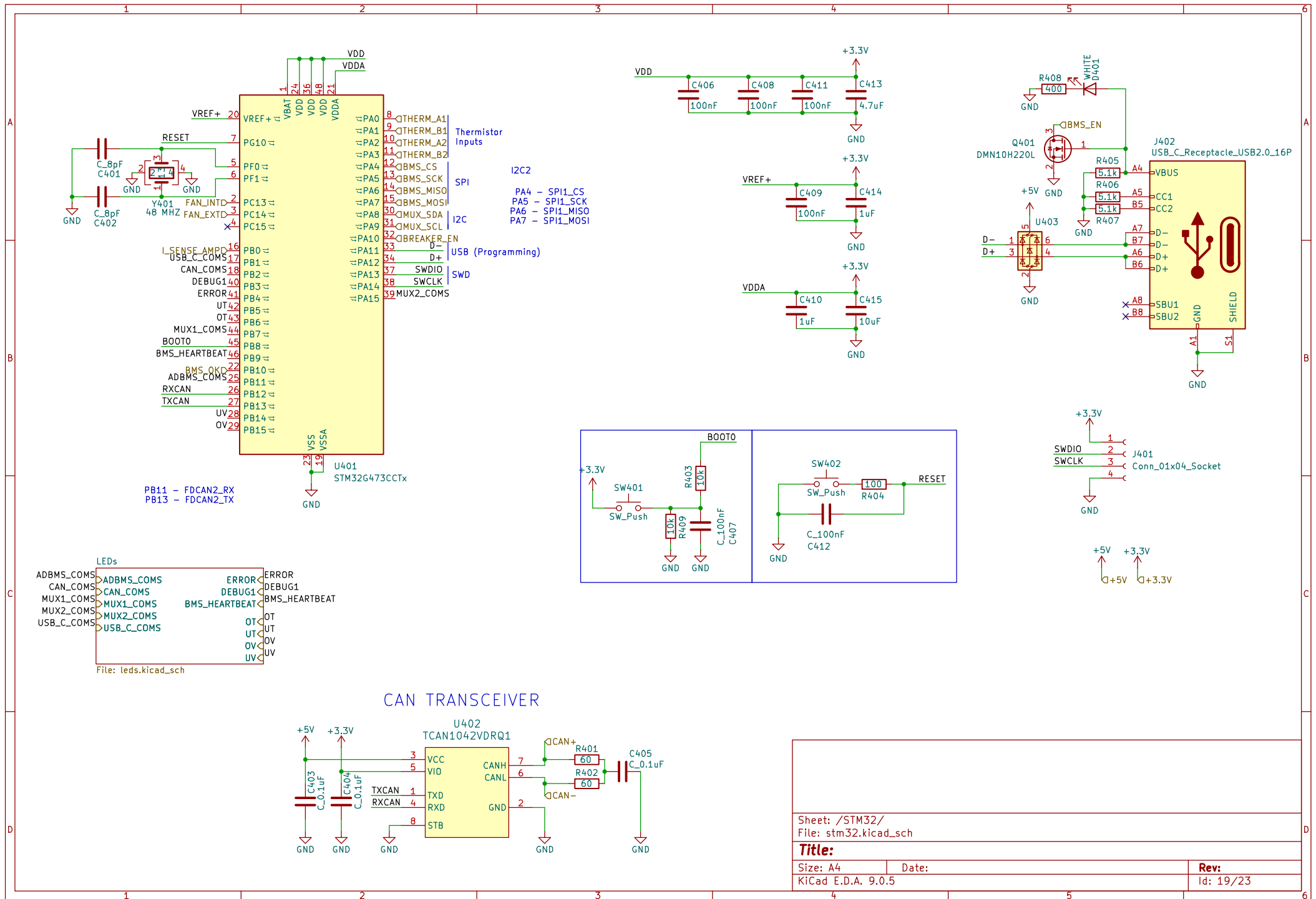
Date:

KiCad E.D.A. 9.0.5

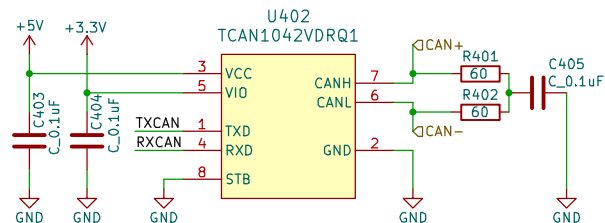
Rev:

Id: 17/23





CAN TRANSCEIVER



Sheet: /STM32/
File: stm32.kicad_sch

Title:

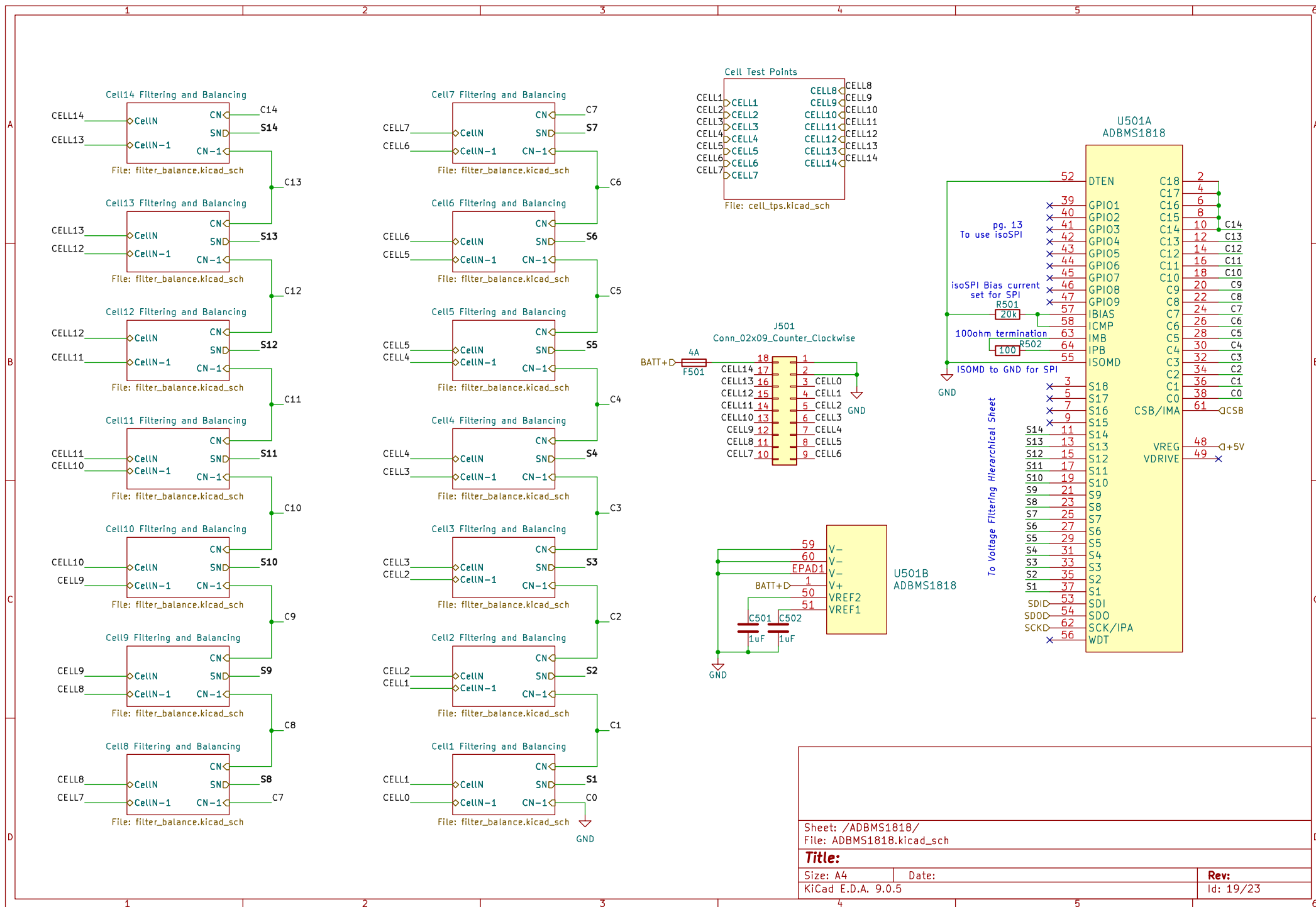
Size: A4

Date:

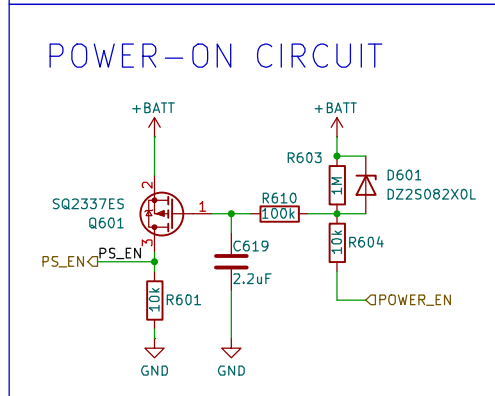
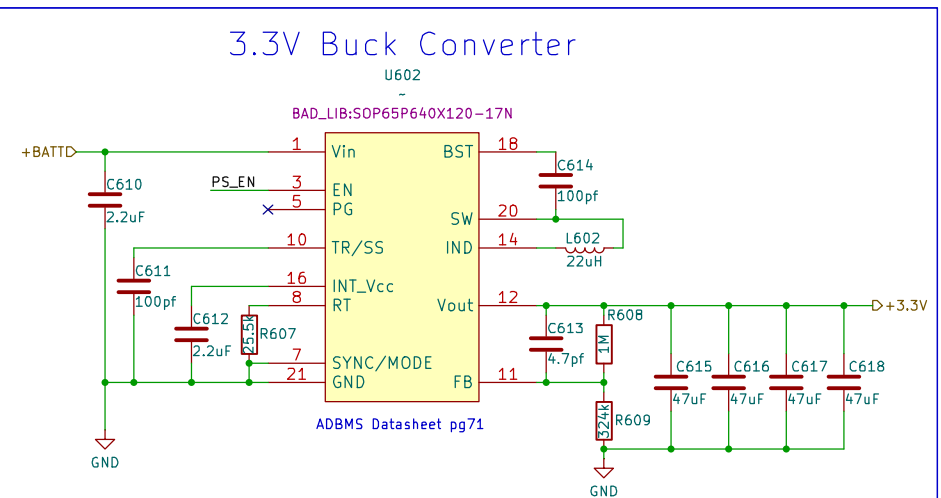
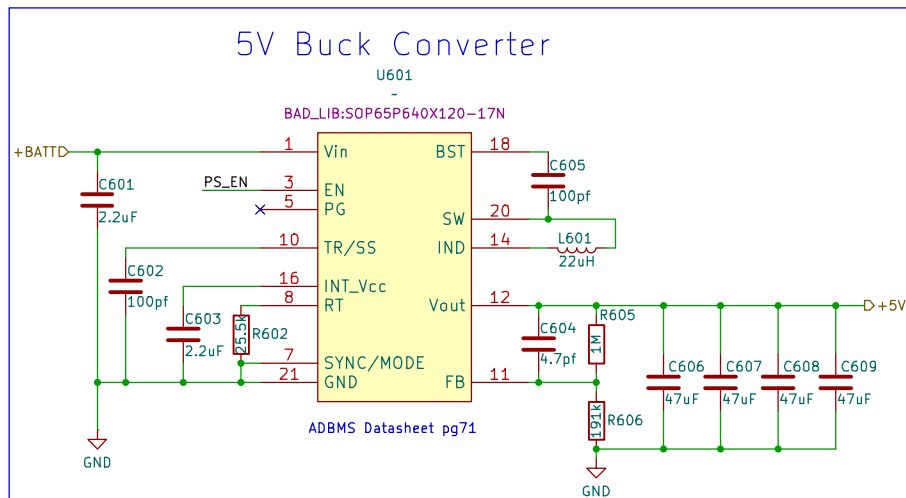
KiCad E.D.A. 9.0.5

Rev:

Id: 19/23



POWER



Sheet: /Power/
File: power.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:

Id: 19/23

This is a Hierarchical Sheet.
That means that this is a "copy-paste"
block for each block shown.
(ie. As seen on "Voltage Filter")

"A cell pin filter of 100 Ω and 10 nF
is recommended for all applications"
ADBMS1818 datasheet pg. 71

However, since it limits the balancing current...

"Note that the antialiasing filter resistor is part of the discharge path
and must be removed or reduced. Use of an RC for added cell
voltage measurement filtering is permitted, but the filter resistor
must remain small, typically around 10 Ω to reduce the effect on the
balance current." pg. 72

Filtering RC circuit as depicted on
ADBMS1818 datasheet pg. 72

Balance Current

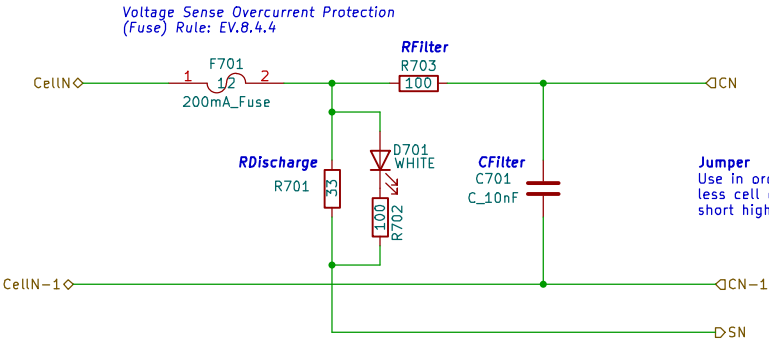
$V = 4.2 \text{ V}$

$R = R_{\text{DISCHARGE}} + R_{\text{FILTER}} + R_{\text{DS_ON}}$
 $= 26 \text{ } \Omega + 10 \text{ } \Omega + 4 \text{ } \Omega$
 $= 40 \text{ } \Omega$

$I = 4.2 \text{ V} / 40 \text{ } \Omega$
 $= 105 \text{ mA}$

Power

$P = I^2 R$
 $= (105 \text{ mA})^2 * 40 \text{ } \Omega$
 $= 0.441 \text{ W}$



Balance current [A]

$= (\% \text{ SOC imbalance} * \text{battery capacity}) / (\text{num of hours to balance})$
 $= (0.05 * 64\text{Ah}) / (20)$
 $= 160 \text{ mA}$

Discharge Resistor [ohms]

$= \text{cell voltage} / \text{balancing current}$
 $= 4.2 \text{ V} / 160\text{mA}$
 $= 26.25 \text{ ohms}$

As long as balance current is below 200mA,
everything is okay to use the internal MOSFETs
of the ADBMS1818 chip for balancing.

This is the calculated balance current for the 10087191NSH5
cells, trying to keep it under 200mA. The example in the
ADBMS1818 datasheet uses 5% SOC imbalance being resolved
in 5 hours, but we can push to a longer balancing time if needed.

Sheet: /ADBMS1818/Cell2 Filtering and Balancing/
File: filter_balance.kicad_sch

Title:

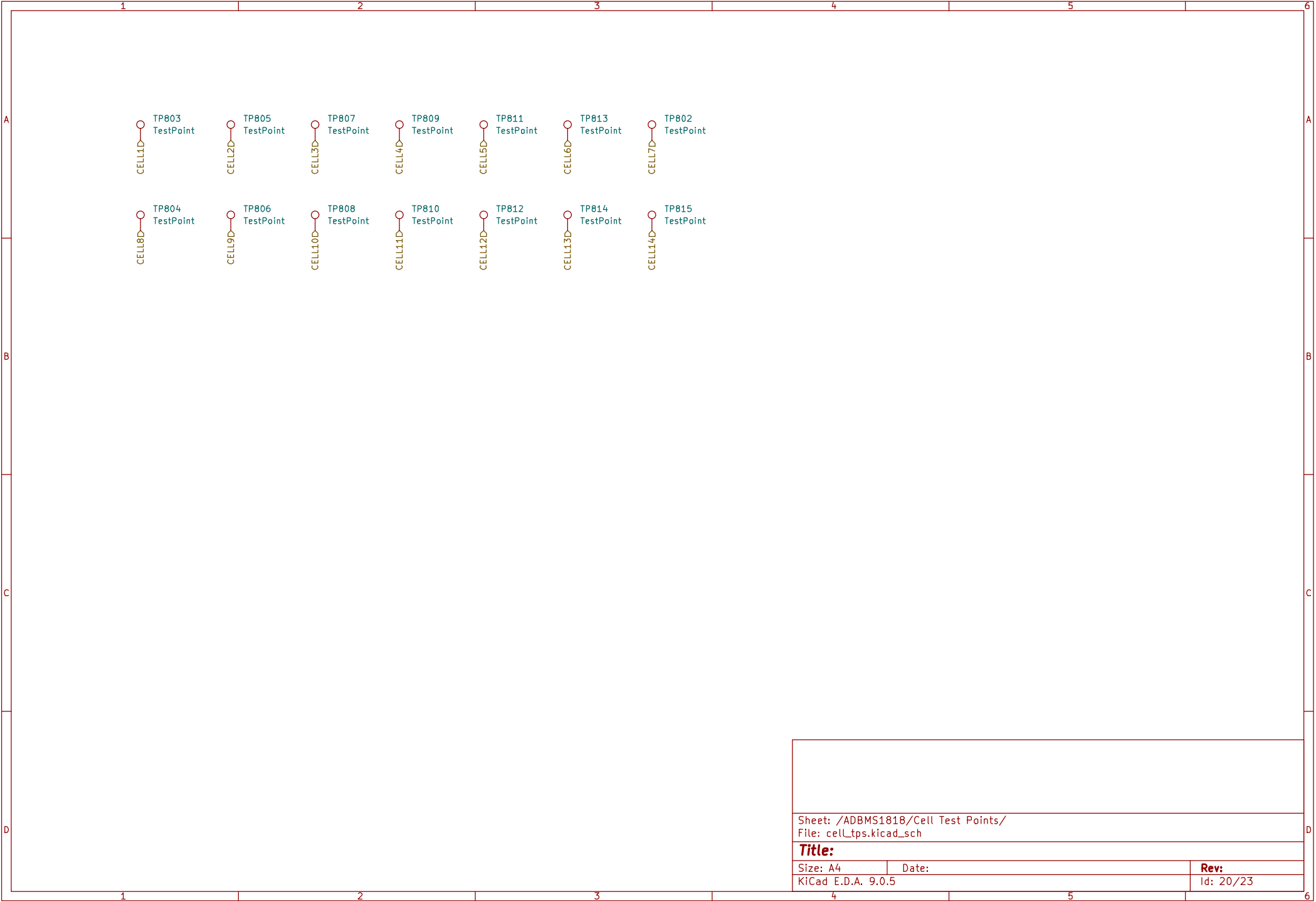
Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:

Id: 20/23



Sheet: /ADBMS1818/Cell Test Points/
File: cell_tps.kicad_sch

Title:

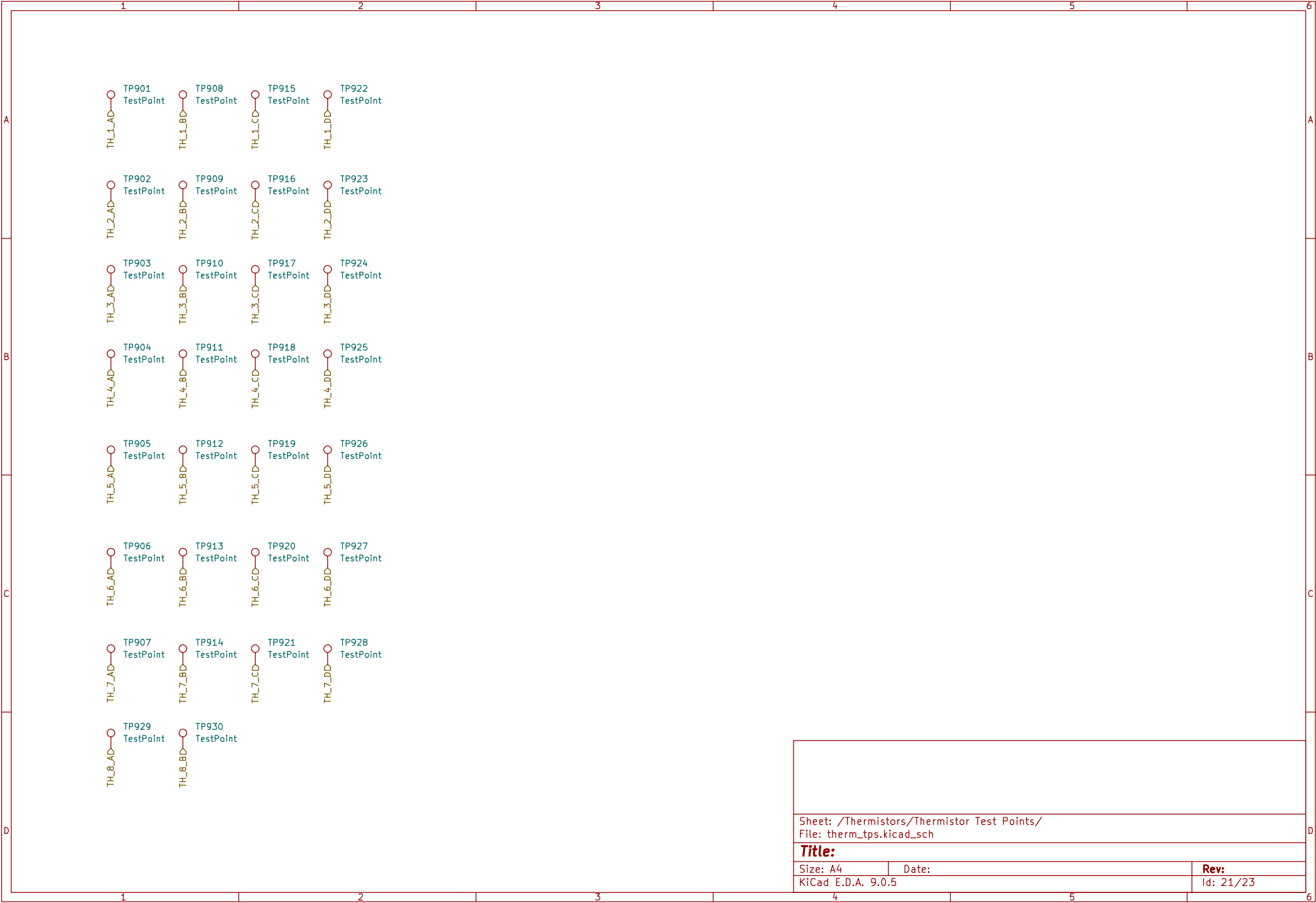
Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:

Id: 20/23



This is a Hierarchical Sheet.
That means that this is a "copy-paste"
block for each block shown.
(ie. As seen on "Voltage Filter")

"A cell pin filter of 100 Ω and 10 nF
is recommended for all applications"
ADBMS1818 datasheet pg. 71

However, since it limits the balancing current...

"Note that the antialiasing filter resistor is part of the discharge path
and must be removed or reduced. Use of an RC for added cell
voltage measurement filtering is permitted, but the filter resistor
must remain small, typically around 10 Ω to reduce the effect on the
balance current." pg. 72

Filtering RC circuit as depicted on
ADBMS1818 datasheet pg. 72

Balance Current

$V = 4.2 \text{ V}$

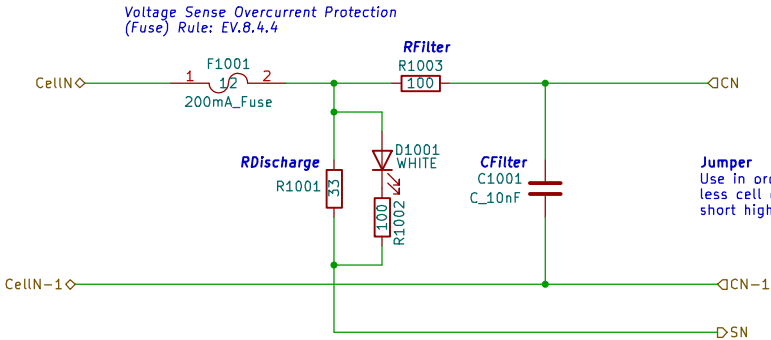
$R = R_{\text{DISCHARGE}} + R_{\text{FILTER}} + R_{\text{DS_ON}}$
 $= 26 \Omega + 10 \Omega + 4 \Omega$
 $= 40 \Omega$

$I = 4.2 \text{ V} / 40 \Omega$
 $= 105 \text{ mA}$

Power

$P = I^2 R$

$= (105 \text{ mA})^2 * 40 \Omega$
 $= 0.441 \text{ W}$



Balance current [A]

$= (\% \text{ SOC imbalance} * \text{battery capacity}) / (\text{num of hours to balance})$

$= (0.05 * 64\text{Ah}) / (20)$

$= 160 \text{ mA}$

Discharge Resistor [ohms]

$= \text{cell voltage} / \text{balancing current}$

$= 4.2 \text{ V} / 160\text{mA}$

$= 26.25 \text{ ohms}$

As long as balance current is below 200mA,
everything is okay to use the internal MOSFETs
of the ADBMS1818 chip for balancing.

This is the calculated balance current for the 10087191NSH5
cells, trying to keep it under 200mA. The example in the
ADBMS1818 datasheet uses 5% SOC imbalance being resolved
in 5 hours, but we can push to a longer balancing time if needed.

Sheet: /ADBMS1818/Cell3 Filtering and Balancing/
File: filter_balance.kicad_sch

Title:

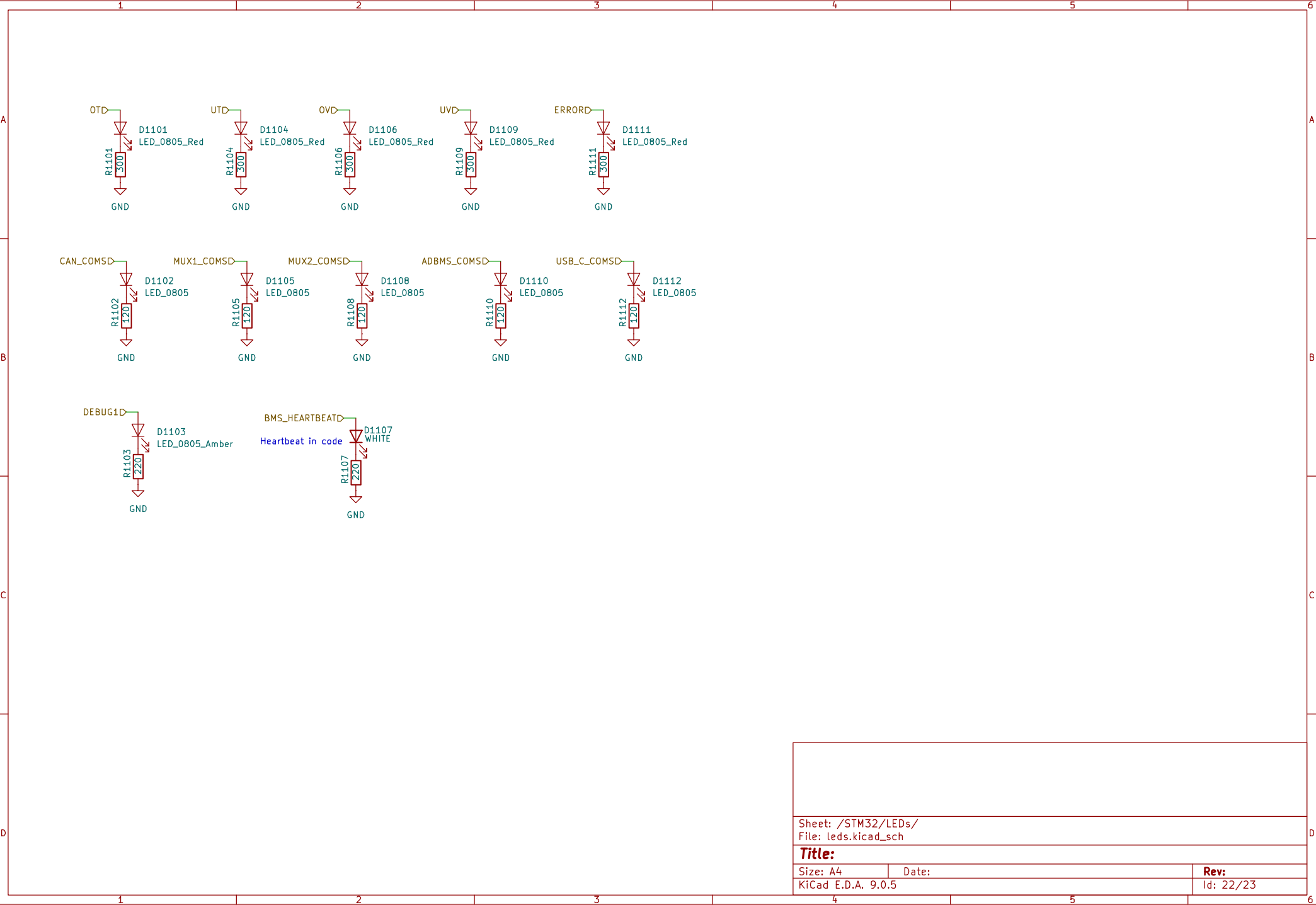
Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:

Id: 21/23



Sheet: /STM32/LEDs/ File: leds.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.5		Id: 22/23

This is a Hierarchical Sheet.
That means that this is a "copy-paste"
block for each block shown.
(ie. As seen on "Voltage Filter")

"A cell pin filter of 100 Ω and 10 nF
is recommended for all applications"
ADBMS1818 datasheet pg. 71

However, since it limits the balancing current...

"Note that the antialiasing filter resistor is part of the discharge path
and must be removed or reduced. Use of an RC for added cell
voltage measurement filtering is permitted, but the filter resistor
must remain small, typically around 10 Ω to reduce the effect on the
balance current." pg. 72

Filtering RC circuit as depicted on
ADBMS1818 datasheet pg. 72

Balance Current

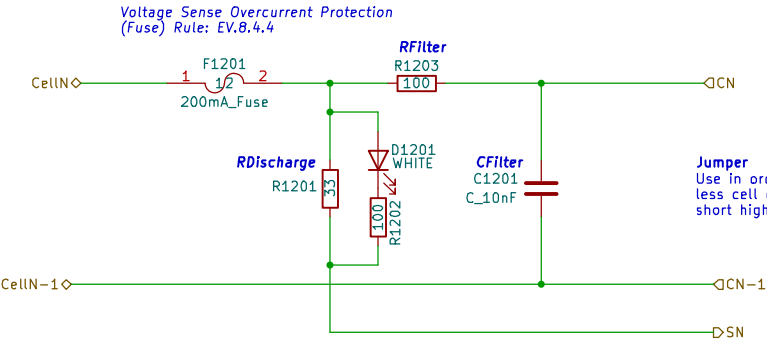
$V = 4.2 \text{ V}$

$R = R_{\text{DISCHARGE}} + R_{\text{FILTER}} + R_{\text{DS_ON}}$
 $= 26 \Omega + 10 \Omega + 4 \Omega$
 $= 40 \Omega$

$I = 4.2 \text{ V} / 40 \Omega$
 $= 105 \text{ mA}$

Power

$P = I^2 R$
 $= (105 \text{ mA})^2 * 40 \Omega$
 $= 0.441 \text{ W}$



Balance current [A]

$= (\% \text{ SOC imbalance} * \text{battery capacity}) / (\text{num of hours to balance})$
 $= (0.05 * 64\text{Ah}) / (20)$
 $= 160 \text{ mA}$

Discharge Resistor [ohms]

$= \text{cell voltage} / \text{balancing current}$
 $= 4.2 \text{ V} / 160\text{mA}$
 $= 26.25 \text{ ohms}$

As long as balance current is below 200mA,
everything is okay to use the internal MOSFETs
of the ADBMS1818 chip for balancing.

This is the calculated balance current for the 10087191NSH5
cells, trying to keep it under 200mA. The example in the
ADBMS1818 datasheet uses 5% SOC imbalance being resolved
in 5 hours, but we can push to a longer balancing time if needed.

Sheet: /ADBMS1818/Cell4 Filtering and Balancing/
File: filter_balance.kicad_sch

Title:

Size: A4

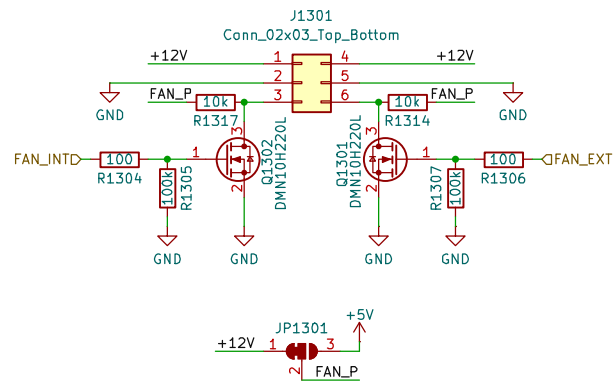
Date:

KiCad E.D.A. 9.0.5

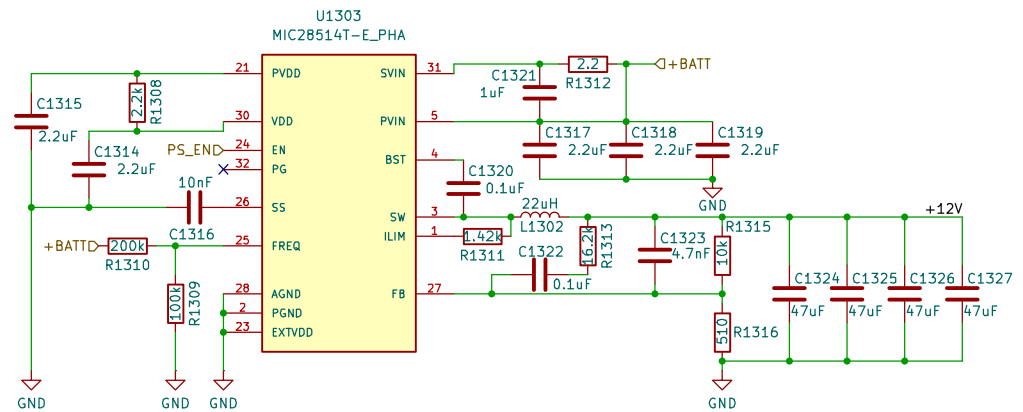
Rev:

Id: 22/23

PWM FETs and Connector



12V Buck Converter



Sheet: /Fans/
File: fans.kicad_sch

Title:

Size: A4
KiCad E.D.A. 9.0.5

Date:

Rev:
Id: 23/23

This is a Hierarchical Sheet.
That means that this is a "copy-paste"
block for each block shown.
(ie. As seen on "Voltage Filter")

"A cell pin filter of 100 Ω and 10 nF
is recommended for all applications"
ADBMS1818 datasheet pg. 71

However, since it limits the balancing current...

"Note that the antialiasing filter resistor is part of the discharge path
and must be removed or reduced. Use of an RC for added cell
voltage measurement filtering is permitted, but the filter resistor
must remain small, typically around 10 Ω to reduce the effect on the
balance current." pg. 72

Filtering RC circuit as depicted on
ADBMS1818 datasheet pg. 72

Balance Current

$V = 4.2 \text{ V}$

$R = R_{\text{DISCHARGE}} + R_{\text{FILTER}} + R_{\text{DS_ON}}$
 $= 26 \text{ } \Omega + 10 \text{ } \Omega + 4 \text{ } \Omega$
 $= 40 \text{ } \Omega$

$I = 4.2 \text{ V} / 40 \text{ } \Omega$
 $= 105 \text{ mA}$

Power

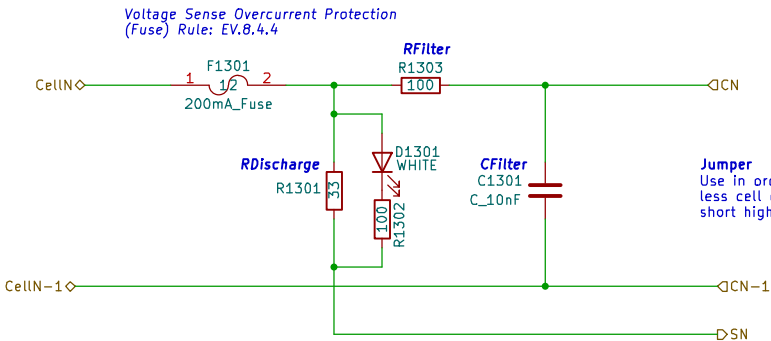
$P = I^2 R$
 $= (105 \text{ mA})^2 * 40 \text{ } \Omega$
 $= 0.441 \text{ W}$

Balance current [A]

$= (\% \text{ SOC imbalance} * \text{battery capacity}) / (\text{num of hours to balance})$
 $= (0.05 * 64\text{Ah}) / (20)$
 $= 160 \text{ mA}$

Discharge Resistor [ohms]

$= \text{cell voltage} / \text{balancing current}$
 $= 4.2 \text{ V} / 160\text{mA}$
 $= 26.25 \text{ ohms}$



Jumper

Use in order to have
less cell connections –
short high rails together

As long as balance current is below 200mA,
everything is okay to use the internal MOSFETs
of the ADBMS1818 chip for balancing.

This is the calculated balance current for the 10087191NSH5
cells, trying to keep it under 200mA. The example in the
ADBMS1818 datasheet uses 5% SOC imbalance being resolved
in 5 hours, but we can push to a longer balancing time if needed.

Sheet: /ADBMS1818/Cell5 Filtering and Balancing/
File: filter_balance.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:

Id: 23/23

This is a Hierarchical Sheet.
That means that this is a "copy-paste"
block for each block shown.
(ie. As seen on "Voltage Filter")

"A cell pin filter of 100 Ω and 10 nF
is recommended for all applications"
ADBMS1818 datasheet pg. 71

However, since it limits the balancing current...

"Note that the antialiasing filter resistor is part of the discharge path
and must be removed or reduced. Use of an RC for added cell
voltage measurement filtering is permitted, but the filter resistor
must remain small, typically around 10 Ω to reduce the effect on the
balance current." pg. 72

Filtering RC circuit as depicted on
ADBMS1818 datasheet pg. 72

Balance Current

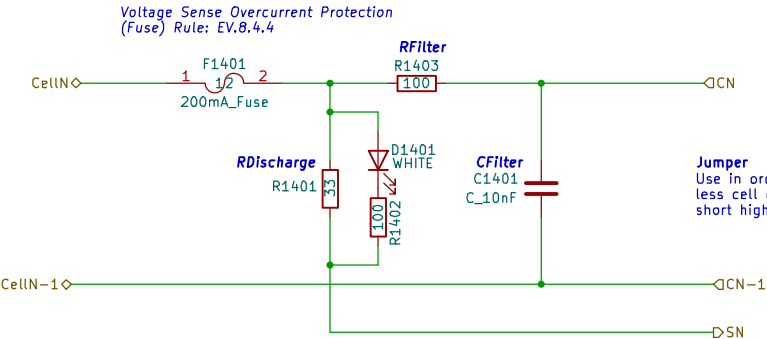
$V = 4.2 \text{ V}$

$R = R_{\text{DISCHARGE}} + R_{\text{FILTER}} + R_{\text{DS_ON}}$
 $= 26 \text{ } \Omega + 10 \text{ } \Omega + 4 \text{ } \Omega$
 $= 40 \text{ } \Omega$

$I = 4.2 \text{ V} / 40 \text{ } \Omega$
 $= 105 \text{ mA}$

Power

$P = I^2 R$
 $= (105 \text{ mA})^2 * 40 \text{ } \Omega$
 $= 0.441 \text{ W}$



Jumper
Use in order to have
less cell connections –
short high rails together

Balance current [A]

$= (\% \text{ SOC imbalance} * \text{battery capacity}) / (\text{num of hours to balance})$
 $= (0.05 * 64\text{Ah}) / (20)$
 $= 160 \text{ mA}$

Discharge Resistor [ohms]

$= \text{cell voltage} / \text{balancing current}$
 $= 4.2 \text{ V} / 160\text{mA}$
 $= 26.25 \text{ ohms}$

As long as balance current is below 200mA,
everything is okay to use the internal MOSFETs
of the ADBMS1818 chip for balancing.

This is the calculated balance current for the 10087191NSH5
cells, trying to keep it under 200mA. The example in the
ADBMS1818 datasheet uses 5% SOC imbalance being resolved
in 5 hours, but we can push to a longer balancing time if needed.

Sheet: /ADBMS1818/Cell6 Filtering and Balancing/
File: filter_balance.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:

Id: 24/23

This is a Hierarchical Sheet.
That means that this is a "copy-paste"
block for each block shown.
(ie. As seen on "Voltage Filter")

"A cell pin filter of 100 Ω and 10 nF
is recommended for all applications"
ADBMS1818 datasheet pg. 71

However, since it limits the balancing current...

"Note that the antialiasing filter resistor is part of the discharge path
and must be removed or reduced. Use of an RC for added cell
voltage measurement filtering is permitted, but the filter resistor
must remain small, typically around 10 Ω to reduce the effect on the
balance current." pg. 72

Filtering RC circuit as depicted on
ADBMS1818 datasheet pg. 72

Balance Current

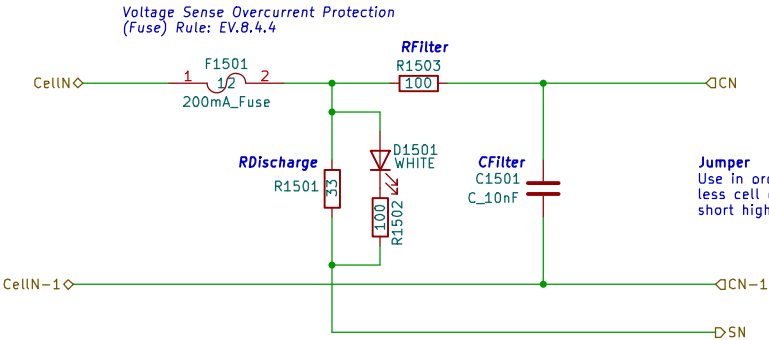
$V = 4.2 \text{ V}$

$R = R_{\text{DISCHARGE}} + R_{\text{FILTER}} + R_{\text{DS_ON}}$
 $= 26 \Omega + 10 \Omega + 4 \Omega$
 $= 40 \Omega$

$I = 4.2 \text{ V} / 40 \Omega$
 $= 105 \text{ mA}$

Power

$P = I^2 R$
 $= (105 \text{ mA})^2 * 40 \Omega$
 $= 0.441 \text{ W}$



Balance current [A]

$= (\% \text{ SOC imbalance} * \text{battery capacity}) / (\text{num of hours to balance})$
 $= (0.05 * 64\text{Ah}) / (20)$
 $= 160 \text{ mA}$

Discharge Resistor [ohms]

$= \text{cell voltage} / \text{balancing current}$
 $= 4.2 \text{ V} / 160\text{mA}$
 $= 26.25 \text{ ohms}$

As long as balance current is below 200mA,
everything is okay to use the internal MOSFETs
of the ADBMS1818 chip for balancing.

This is the calculated balance current for the 10087191NSH5
cells, trying to keep it under 200mA. The example in the
ADBMS1818 datasheet uses 5% SOC imbalance being resolved
in 5 hours, but we can push to a longer balancing time if needed.

Sheet: /ADBMS1818/Cell7 Filtering and Balancing/
File: filter_balance.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:

Id: 25/23

This is a Hierarchical Sheet.
That means that this is a "copy-paste"
block for each block shown.
(ie. As seen on "Voltage Filter")

"A cell pin filter of 100 Ω and 10 nF
is recommended for all applications"
ADBMS1818 datasheet pg. 71

However, since it limits the balancing current...

"Note that the antialiasing filter resistor is part of the discharge path
and must be removed or reduced. Use of an RC for added cell
voltage measurement filtering is permitted, but the filter resistor
must remain small, typically around 10 Ω to reduce the effect on the
balance current." pg. 72

Filtering RC circuit as depicted on
ADBMS1818 datasheet pg. 72

Balance Current

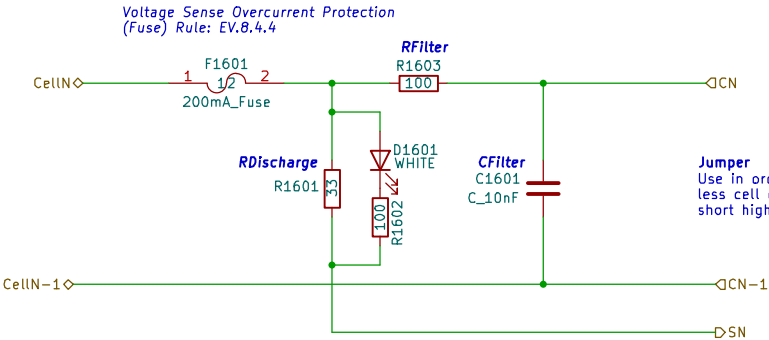
$V = 4.2 \text{ V}$

$R = R_{\text{DISCHARGE}} + R_{\text{FILTER}} + R_{\text{DS_ON}}$
 $= 26 \text{ } \Omega + 10 \text{ } \Omega + 4 \text{ } \Omega$
 $= 40 \text{ } \Omega$

$I = 4.2 \text{ V} / 40 \text{ } \Omega$
 $= 105 \text{ mA}$

Power

$P = I^2 R$
 $= (105 \text{ mA})^2 * 40 \text{ } \Omega$
 $= 0.441 \text{ W}$



Jumper

Use in order to have
less cell connections –
short high rails together

Balance current [A]

$= (\% \text{ SOC imbalance} * \text{battery capacity}) / (\text{num of hours to balance})$
 $= (0.05 * 64\text{Ah}) / (20)$
 $= 160 \text{ mA}$

Discharge Resistor [ohms]

$= \text{cell voltage} / \text{balancing current}$
 $= 4.2 \text{ V} / 160\text{mA}$
 $= 26.25 \text{ ohms}$

As long as balance current is below 200mA,
everything is okay to use the internal MOSFETs
of the ADBMS1818 chip for balancing.

This is the calculated balance current for the 10087191NSH5
cells, trying to keep it under 200mA. The example in the
ADBMS1818 datasheet uses 5% SOC imbalance being resolved
in 5 hours, but we can push to a longer balancing time if needed.

Sheet: /ADBMS1818/Cell8 Filtering and Balancing/
File: filter_balance.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:

Id: 26/23

This is a Hierarchical Sheet.
That means that this is a "copy-paste"
block for each block shown.
(ie. As seen on "Voltage Filter")

"A cell pin filter of 100 Ω and 10 nF
is recommended for all applications"
ADBMS1818 datasheet pg. 71

However, since it limits the balancing current...

"Note that the antialiasing filter resistor is part of the discharge path
and must be removed or reduced. Use of an RC for added cell
voltage measurement filtering is permitted, but the filter resistor
must remain small, typically around 10 Ω to reduce the effect on the
balance current." pg. 72

Filtering RC circuit as depicted on
ADBMS1818 datasheet pg. 72

Voltage Sense Overcurrent Protection
(Fuse) Rule: EV.8.4.4

Balance Current

$V = 4.2 \text{ V}$

$R = R_{\text{DISCHARGE}} + R_{\text{FILTER}} + R_{\text{DS_ON}}$
 $= 26 \Omega + 10 \Omega + 4 \Omega$
 $= 40 \Omega$

$I = 4.2 \text{ V} / 40 \Omega$
 $= 105 \text{ mA}$

Power

$P = I^2 R$
 $= (105 \text{ mA})^2 * 40 \Omega$
 $= 0.441 \text{ W}$

CellIN

F1701
200mA_Fuse

RFilter
R1703
100

RDIscharge
R1701
33

D1701
WHITE

CFilter
C1701
C_10nF

Jumper
Use in order to have
less cell connections –
short high rails together

CellIN-1

ICN

ICN-1

DSN

As long as balance current is below 200mA,
everything is okay to use the internal MOSFETs
of the ADBMS1818 chip for balancing.

This is the calculated balance current for the 10087191NSH5
cells, trying to keep it under 200mA. The example in the
ADBMS1818 datasheet uses 5% SOC imbalance being resolved
in 5 hours, but we can push to a longer balancing time if needed.

Balance current [A]

$= (\% \text{ SOC Imbalance} * \text{battery capacity}) / (\text{num of hours to balance})$

$= (0.05 * 64\text{Ah}) / (20)$

$= 160 \text{ mA}$

Discharge Resistor [ohms]

$= \text{cell voltage} / \text{balancing current}$

$= 4.2 \text{ V} / 160\text{mA}$

$= 26.25 \text{ ohms}$

Sheet: /ADBMS1818/Cell9 Filtering and Balancing/
File: filter_balance.kicad_sch

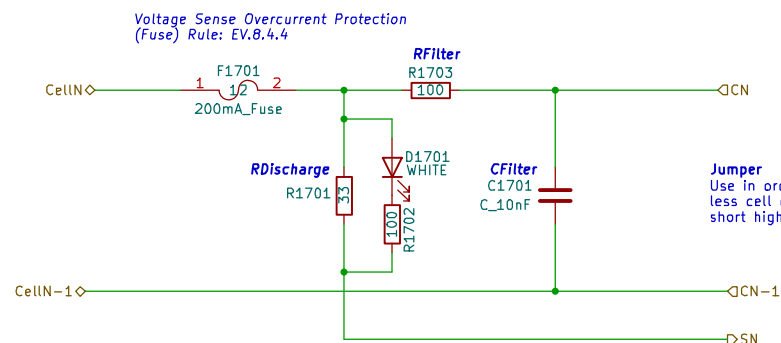
Title:

Size: A4 Date: Rev:

KiCad E.D.A. 9.0.5 Id: 27/23

However, since it limits the balancing current...

Filtering RC circuit as depicted on
ADBMS1818 datasheet pg. 72


$$V = 4,2 \text{ V}$$

$$I = 4.2 \text{ V} / 40 \text{ } \Omega$$
$$= 105 \text{ mA}$$

Power

$$\begin{aligned} P &= I^2 R \\ &= (105 \text{ mA})^2 \cdot 40 \, \Omega \\ &= 0.441 \text{ W} \end{aligned}$$

Balance current [A]

$$\begin{aligned} &= (\% \text{ SOC imbalance} * \text{battery capacity}) / (\text{num of hours to balance}) \\ &= (0.05 * 64\text{Ah}) / (20) \\ &= 160 \text{ mA} \end{aligned}$$

Discharge Resistor [ohms]

$$\begin{aligned} &= \text{cell voltage} / \text{balancing current} \\ &= 4.2 \text{ V} / 160\text{mA} \\ &= 26.25 \text{ ohms} \end{aligned}$$

As long as balance current is below 200mA, everything is okay to use the internal MOSFETs of the ADBMS1818 chip for balancing.

This is the calculated balance current for the 10087191NSH5 cells, trying to keep it under 200mA. The example in the ADBMS1818 datasheet uses 5% SOC imbalance being resolved in 5 hours, but we can push to a longer balancing time if needed.

Sheet: /ADBMS1818/Cell9 Filtering and Balancing/
File: filter_balance.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:	
------	--

Id: 27/23

This is a Hierarchical Sheet.
That means that this is a "copy-paste"
block for each block shown.
(ie. As seen on "Voltage Filter")

"A cell pin filter of 100 Ω and 10 nF
is recommended for all applications"
ADBMS1818 datasheet pg. 71

However, since it limits the balancing current...

"Note that the antialiasing filter resistor is part of the discharge path
and must be removed or reduced. Use of an RC for added cell
voltage measurement filtering is permitted, but the filter resistor
must remain small, typically around 10 Ω to reduce the effect on the
balance current." pg. 72

Filtering RC circuit as depicted on
ADBMS1818 datasheet pg. 72

Balance Current

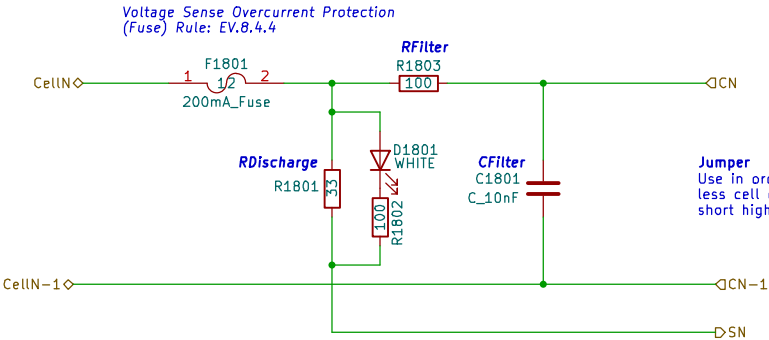
$V = 4.2 \text{ V}$

$R = R_{\text{DISCHARGE}} + R_{\text{FILTER}} + R_{\text{DS_ON}}$
 $= 26 \Omega + 10 \Omega + 4 \Omega$
 $= 40 \Omega$

$I = 4.2 \text{ V} / 40 \Omega$
 $= 105 \text{ mA}$

Power

$P = I^2 R$
 $= (105 \text{ mA})^2 * 40 \Omega$
 $= 0.441 \text{ W}$



Balance current [A]

$= (\% \text{ SOC imbalance} * \text{battery capacity}) / (\text{num of hours to balance})$
 $= (0.05 * 64\text{Ah}) / (20)$
 $= 160 \text{ mA}$

Discharge Resistor [ohms]

$= \text{cell voltage} / \text{balancing current}$
 $= 4.2 \text{ V} / 160\text{mA}$
 $= 26.25 \text{ ohms}$

As long as balance current is below 200mA,
everything is okay to use the internal MOSFETs
of the ADBMS1818 chip for balancing.

This is the calculated balance current for the 10087191NSH5
cells, trying to keep it under 200mA. The example in the
ADBMS1818 datasheet uses 5% SOC imbalance being resolved
in 5 hours, but we can push to a longer balancing time if needed.

Sheet: /ADBMS1818/Cell10 Filtering and Balancing/
File: filter_balance.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:

Id: 28/23

This is a Hierarchical Sheet.
That means that this is a "copy-paste"
block for each block shown.
(ie. As seen on "Voltage Filter")

"A cell pin filter of 100 Ω and 10 nF
is recommended for all applications"
ADBMS1818 datasheet pg. 71

However, since it limits the balancing current...

"Note that the antialiasing filter resistor is part of the discharge path
and must be removed or reduced. Use of an RC for added cell
voltage measurement filtering is permitted, but the filter resistor
must remain small, typically around 10 Ω to reduce the effect on the
balance current." pg. 72

Filtering RC circuit as depicted on
ADBMS1818 datasheet pg. 72

Balance Current

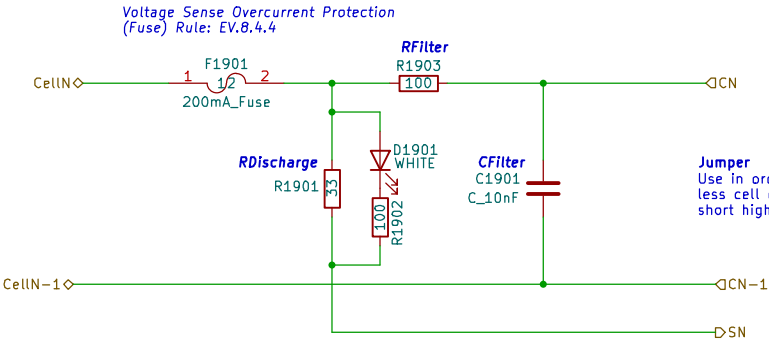
$V = 4.2\text{ V}$

$R = R_{\text{DISCHARGE}} + R_{\text{FILTER}} + R_{\text{DS_ON}}$
 $= 26\ \Omega + 10\ \Omega + 4\ \Omega$
 $= 40\ \Omega$

$I = 4.2\text{ V} / 40\ \Omega$
 $= 105\text{ mA}$

Power

$P = I^2 R$
 $= (105\text{ mA})^2 \cdot 40\ \Omega$
 $= 0.441\text{ W}$



Balance current [A]

$= (\% \text{ SOC imbalance} \cdot \text{battery capacity}) / (\text{num of hours to balance})$
 $= (0.05 \cdot 64\text{Ah}) / (20)$
 $= 160\text{ mA}$

Discharge Resistor [ohms]

$= \text{cell voltage} / \text{balancing current}$
 $= 4.2\text{ V} / 160\text{mA}$
 $= 26.25\text{ ohms}$

As long as balance current is below 200mA,
everything is okay to use the internal MOSFETs
of the ADBMS1818 chip for balancing.

This is the calculated balance current for the 10087191NSH5
cells, trying to keep it under 200mA. The example in the
ADBMS1818 datasheet uses 5% SOC imbalance being resolved
in 5 hours, but we can push to a longer balancing time if needed.

Sheet: /ADBMS1818/Cell11 Filtering and Balancing/
File: filter_balance.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:

Id: 29/23

This is a Hierarchical Sheet.
That means that this is a "copy-paste"
block for each block shown.
(ie. As seen on "Voltage Filter")

"A cell pin filter of 100 Ω and 10 nF
is recommended for all applications"
ADBMS1818 datasheet pg. 71

However, since it limits the balancing current...

"Note that the antialiasing filter resistor is part of the discharge path
and must be removed or reduced. Use of an RC for added cell
voltage measurement filtering is permitted, but the filter resistor
must remain small, typically around 10 Ω to reduce the effect on the
balance current." pg. 72

Filtering RC circuit as depicted on
ADBMS1818 datasheet pg. 72

Balance Current

$V = 4.2 \text{ V}$

$R = R_{\text{DISCHARGE}} + R_{\text{FILTER}} + R_{\text{DS_ON}}$
 $= 26 \Omega + 10 \Omega + 4 \Omega$
 $= 40 \Omega$

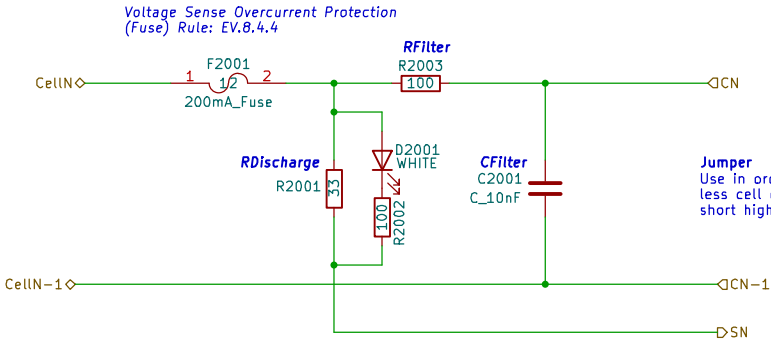
$I = 4.2 \text{ V} / 40 \Omega$
 $= 105 \text{ mA}$

Power

$P = I^2 R$

$= (105 \text{ mA})^2 \cdot 40 \Omega$

$= 0.441 \text{ W}$



Balance current [A]

$= (\% \text{ SOC imbalance} \cdot \text{battery capacity}) / (\text{num of hours to balance})$

$= (0.05 \cdot 64\text{Ah}) / (20)$

$= 160 \text{ mA}$

Discharge Resistor [ohms]

$= \text{cell voltage} / \text{balancing current}$

$= 4.2 \text{ V} / 160\text{mA}$

$= 26.25 \text{ ohms}$

As long as balance current is below 200mA,
everything is okay to use the internal MOSFETs
of the ADBMS1818 chip for balancing.

This is the calculated balance current for the 10087191NSH5
cells, trying to keep it under 200mA. The example in the
ADBMS1818 datasheet uses 5% SOC imbalance being resolved
in 5 hours, but we can push to a longer balancing time if needed.

Sheet: /ADBMS1818/Cell12 Filtering and Balancing/
File: filter_balance.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:

Id: 30/23

This is a Hierarchical Sheet.
That means that this is a "copy-paste"
block for each block shown.
(ie. As seen on "Voltage Filter")

"A cell pin filter of 100 Ω and 10 nF
is recommended for all applications"
ADBMS1818 datasheet pg. 71

However, since it limits the balancing current...

"Note that the antialiasing filter resistor is part of the discharge path
and must be removed or reduced. Use of an RC for added cell
voltage measurement filtering is permitted, but the filter resistor
must remain small, typically around 10 Ω to reduce the effect on the
balance current." pg. 72

Filtering RC circuit as depicted on
ADBMS1818 datasheet pg. 72

Balance Current

$V = 4.2 \text{ V}$

$R = R_{\text{DISCHARGE}} + R_{\text{FILTER}} + R_{\text{DS_ON}}$
 $= 26 \Omega + 10 \Omega + 4 \Omega$
 $= 40 \Omega$

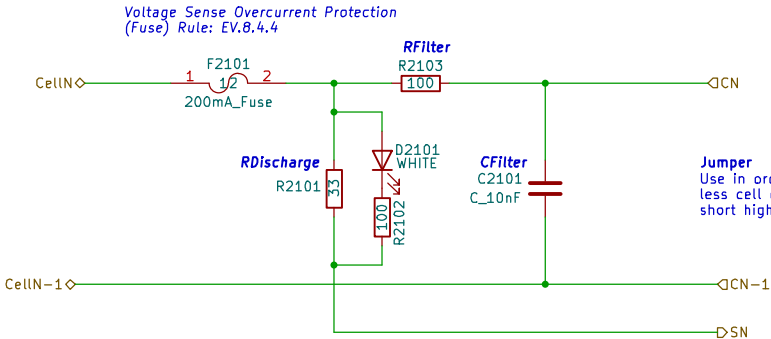
$I = 4.2 \text{ V} / 40 \Omega$
 $= 105 \text{ mA}$

Power

$P = I^2 R$

$= (105 \text{ mA})^2 \cdot 40 \Omega$

$= 0.441 \text{ W}$



Balance current [A]

$= (\% \text{ SOC imbalance} \cdot \text{battery capacity}) / (\text{num of hours to balance})$

$= (0.05 \cdot 64\text{Ah}) / (20)$

$= 160 \text{ mA}$

Discharge Resistor [ohms]

$= \text{cell voltage} / \text{balancing current}$

$= 4.2 \text{ V} / 160\text{mA}$

$= 26.25 \text{ ohms}$

As long as balance current is below 200mA,
everything is okay to use the internal MOSFETs
of the ADBMS1818 chip for balancing.

This is the calculated balance current for the 10087191NSH5
cells, trying to keep it under 200mA. The example in the
ADBMS1818 datasheet uses 5% SOC imbalance being resolved
in 5 hours, but we can push to a longer balancing time if needed.

Sheet: /ADBMS1818/Cell13 Filtering and Balancing/
File: filter_balance.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:

Id: 31/23

This is a Hierarchical Sheet.
That means that this is a "copy-paste"
block for each block shown.
(ie. As seen on "Voltage Filter")

"A cell pin filter of 100 Ω and 10 nF
is recommended for all applications"
ADBMS1818 datasheet pg. 71

However, since it limits the balancing current...

"Note that the antialiasing filter resistor is part of the discharge path
and must be removed or reduced. Use of an RC for added cell
voltage measurement filtering is permitted, but the filter resistor
must remain small, typically around 10 Ω to reduce the effect on the
balance current." pg. 72

Filtering RC circuit as depicted on
ADBMS1818 datasheet pg. 72

Balance Current

$V = 4.2 \text{ V}$

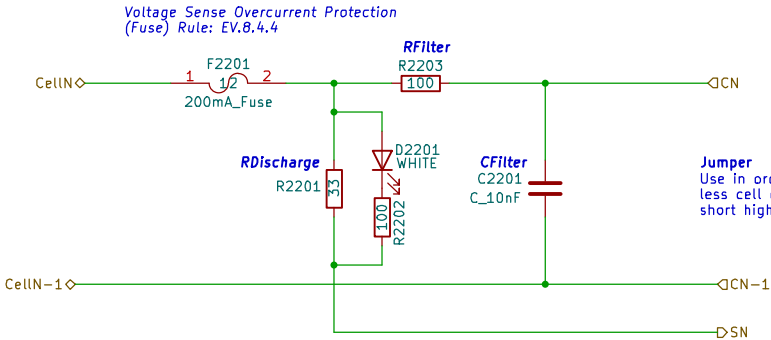
$R = R_{\text{DISCHARGE}} + R_{\text{FILTER}} + R_{\text{DS_ON}}$
 $= 26 \Omega + 10 \Omega + 4 \Omega$
 $= 40 \Omega$

$I = 4.2 \text{ V} / 40 \Omega$
 $= 105 \text{ mA}$

Power

$P = I^2 R$

$= (105 \text{ mA})^2 \cdot 40 \Omega$
 $= 0.441 \text{ W}$



Balance current [A]

$= (\% \text{ SOC imbalance} \cdot \text{battery capacity}) / (\text{num of hours to balance})$

$= (0.05 \cdot 64\text{Ah}) / (20)$

$= 160 \text{ mA}$

Discharge Resistor [ohms]

$= \text{cell voltage} / \text{balancing current}$

$= 4.2 \text{ V} / 160\text{mA}$

$= 26.25 \text{ ohms}$

As long as balance current is below 200mA,
everything is okay to use the internal MOSFETs
of the ADBMS1818 chip for balancing.

This is the calculated balance current for the 10087191NSH5
cells, trying to keep it under 200mA. The example in the
ADBMS1818 datasheet uses 5% SOC imbalance being resolved
in 5 hours, but we can push to a longer balancing time if needed.

Sheet: /ADBMS1818/Cell14 Filtering and Balancing/
File: filter_balance.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:

Id: 32/23